

ASSIGNMENT BRIEF

HTU Course No: 40201391	HTU Course Name: Computing Research Project
BTEC Unit Code: K/618/7425	BTEC UNIT Name: Computing Research Project



Student Name/ID Number/Section	
HTU Course Number and Title	40201391 Computing Research Project
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Academic Year	2025-2026 1
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Assignment Title	Cybersecurity research
Assignment Ref No	1
Issue Date	19/10/2025
Formative Assessment dates	From 22/10/2025 to 15/01/2026
Submission Date	31/01/2026
IV Name & Date	Samer Suleiman 18/10/2025
Submission Format	
The submission for this assignment is: 1. An individual written detailed report that answered the questions illustrated in the Assignment Brief and Guidance section. 2. A presentation for the report content. 3. Research Ethics Form. 4. The primary and secondary data in csv files if possible. 5. The files include the code, where applicable. Report guidelines: In your report, you should make use of headings, paragraphs, and subsections as appropriate. Your report should be: 1. In a form of a word file submitted via the elearning. 2. Written in a formal business style using Times New Roman and font size 12. You are required to submit the declaration form to confirm that the submitted work is entirely your own and that you fully understand the consequences of plagiarism.	
Unit Learning Outcomes	

LO1 Examine appropriate research methodologies and approaches as part of the research process

LO2 Conduct and analyse research relevant to a computing research project

LO3 Communicate the outcomes of a research project to identified stakeholders

LO4 Reflect on the application of research methodologies and concepts

Assignment Brief and Guidance

Part 1: Research Proposal Development

You are required to construct a research proposal that defines a clear research title, research question(s), and/or research hypothesis related to one of the following cybersecurity subtopics:

- Insider threats and risks from employees or partners
- The evolution of cybercrime and trends in cyberattacks and criminal tactics
- Supply chain cybersecurity risks from third-party software and vendors
- Mobile security and data protection on smartphones and tablets
- Deepfake technology and threats from manipulated media
- Securing critical infrastructure systems from cyberattacks
- Zero trust security models and their role in strengthening cybersecurity
- Dark web activities and the mitigation of illegal online markets
- Quantum computing and its implications for cybersecurity
- Cybersecurity in virtualized networks
- The evolution of legislation to address current and future cyber threats

Your proposal must include the following sections:

1. Title Page
2. Introduction

- Background of the topic
- Introduction to the research area

- Literature Review
- Research Objectives, Research Questions, and/or Research Hypothesis
- Methodology
 - Identify whether your study uses a qualitative, quantitative, or mixed-method approach
 - Evaluate common research methods (e.g., survey, interview, experiment, grounded theory, etc.)
 - Justify the selected method based on the philosophical research position
 - Critically assess the suitability of the selected methodology for your computing research project
- Methodology Flowchart
 - Design a flowchart that summarizes your proposed research methodology

Part 2: Data Collection Planning and Ethical Considerations

You must perform both primary and secondary research related to your selected topic. Before collecting the data, you are required to:

- Discuss the quality, reliability, and limitations of your selected data collection techniques
- Identify and analyze potential access, privacy, and ethical issues
- Evaluate these issues in relation to your selected research strategy

Part 3: Research Implementation, Analysis, and Findings

At this stage, you will conduct the actual research by completing the following:

1. Data Collection
 - Collect the required data to address your research question(s)

- Clearly describe your data collection tool (e.g., survey, interview, experiment, etc.)

- Data Analysis

- Apply appropriate analysis tools and techniques
- Examine and interpret the results and findings

- Discussion and Recommendations

- Explain how the results address your research objectives
- Clearly state the answers to your research questions
- Present well-justified recommendations based on your findings

Part 4: Reflection and Future Improvements

You are required to critically reflect on the entire research process by addressing the following:

- Evaluate the effectiveness of the selected research methods in achieving the research aims
- Discuss alternative research methodologies that could have been used
- Reflect on the skills and experience gained during the research project
- Based on lessons learned, propose future improvements and further research considerations
- Justify your recommendations based on your personal reflection

Learning Outcomes and Assessment Criteria			
Learning Outcome	Pass	Merit	Distinction
LO1 Examine appropriate research methodologies and approaches as part of the research process	P1 Produce a research proposal that clearly defines a research question or hypothesis supported by a literature review. P2 Examine appropriate research methods and approaches to primary and secondary research.	M1 Analyze different research approaches and methodology and make justifications for the choice of methods selected based on philosophical/theoretical frameworks.	D1 Critically evaluate research methodologies and processes in application to a computing research project to justify chosen research methods and analysis.
	P3 Conduct primary and secondary research using appropriate methods for a computing research project that consider costs, access and ethical issues. P4 Apply appropriate analytical tools analyse research findings and data.		
LO2 Conduct and analyse research relevant to a computing research project	P5 Communicate research outcomes in an appropriate manner for the intended audience.	M2 Discuss merits, limitations and pitfalls of approaches to data collection and analysis.	D2 Evaluate outcomes and make valid, justified recommendations.
LO3 Communicate the outcomes of a research project to identified stakeholders	P6 Discuss the effectiveness of research methods applied for meeting objectives of the computing research project. P7 Discuss alternative research methodologies and lessons learnt in view of the outcomes.	M3 Analyse the extent to which outcomes meet set research objectives and communicate judgements effectively for the intended audience.	D3 Demonstrate reflection and engagement in the resource process leading to recommended actions for future improvement.
LO4 Reflect on the application of research methodologies and concepts		M4 Analyse results in recommended actions for improvements and future research considerations.	